

# Package ‘ttsapi’

January 12, 2026

**Title** OpenAI-Compatible Text-to-Speech API Client

**Version** 0.1.0

**Description** An R client for Text-to-Speech APIs supporting multiple backends:  
OpenAI-compatible APIs (OpenAI, Chatterbox, LM Studio, OpenWebUI, AnythingLLM)  
and ElevenLabs. Provides a unified interface for speech synthesis with  
backend-specific parameters.

**License** MIT + file LICENSE

**Encoding** UTF-8

**URL** <https://github.com/cornball-ai/ttsapi>

**BugReports** <https://github.com/cornball-ai/ttsapi/issues>

**Imports** curl, jsonlite

**Suggests** processx, gputcl, tinytest

**NeedsCompilation** no

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|                                  |                                            |
|----------------------------------|--------------------------------------------|
| <code>.apply_speed_ffmpeg</code> | <i>Apply speed adjustment using ffmpeg</i> |
|----------------------------------|--------------------------------------------|

---

**Description**

Apply speed adjustment using ffmpeg

**Usage**

`.apply_speed_ffmpeg(input_file, output_file, speed)`

---

|                                     |                              |
|-------------------------------------|------------------------------|
| <code>.fetch_voices_from_api</code> | <i>Fetch voices from API</i> |
|-------------------------------------|------------------------------|

---

**Description**

Fetch voices from API

**Usage**

`.fetch_voices_from_api(base_url, timeout)`

---

|                      |                      |
|----------------------|----------------------|
| <code>.onLoad</code> | <code>.onLoad</code> |
|----------------------|----------------------|

---

### Description

`.onLoad`

### Usage

`.onLoad(libname, pkgname)`

---

|                                   |                                     |
|-----------------------------------|-------------------------------------|
| <code>.try_health_endpoint</code> | <i>Try a single health endpoint</i> |
|-----------------------------------|-------------------------------------|

---

### Description

Try a single health endpoint

### Usage

`.try_health_endpoint(base, endpoint, timeout)`

---

|                              |                               |
|------------------------------|-------------------------------|
| <code>.tts_elevenlabs</code> | <i>ElevenLabs TTS backend</i> |
|------------------------------|-------------------------------|

---

### Description

ElevenLabs TTS backend

### Usage

`.tts_elevenlabs(input, voice_id, file, model, stability, similarity_boost)`

---

|                       |                             |
|-----------------------|-----------------------------|
| <code>.tts_get</code> | <i>GET from the TTS API</i> |
|-----------------------|-----------------------------|

---

**Description**

GET from the TTS API

**Usage**

```
.tts_get(endpoint)
```

**Arguments**

|          |                              |
|----------|------------------------------|
| endpoint | Character. The API endpoint. |
|----------|------------------------------|

**Value**

Parsed JSON response.

---

|                                |                                 |
|--------------------------------|---------------------------------|
| <code>.tts_get_api_base</code> | <i>Get the TTS API Base URL</i> |
|--------------------------------|---------------------------------|

---

**Description**

Get the TTS API Base URL

**Usage**

```
.tts_get_api_base()
```

**Value**

Character string with the API base URL.

---

|                  |                            |
|------------------|----------------------------|
| .tts_get_api_key | <i>Get the TTS API Key</i> |
|------------------|----------------------------|

---

### Description

Get the TTS API Key

### Usage

```
.tts_get_api_key()
```

### Value

Character string with the API key, or NULL if not set.

---

|                |                                 |
|----------------|---------------------------------|
| .tts_post_json | <i>POST JSON to the TTS API</i> |
|----------------|---------------------------------|

---

### Description

POST JSON to the TTS API

### Usage

```
.tts_post_json(endpoint, body, expect_binary)
```

### Arguments

|               |                                           |
|---------------|-------------------------------------------|
| endpoint      | Character. The API endpoint.              |
| body          | List. Request body.                       |
| expect_binary | Logical. If TRUE, expect binary response. |

### Value

Response content.

---

|                           |                                            |
|---------------------------|--------------------------------------------|
| <code>.tts_request</code> | <i>Make an HTTP Request to the TTS API</i> |
|---------------------------|--------------------------------------------|

---

**Description**

Low-level HTTP wrapper around `curl::curl_fetch_memory()`.

**Usage**

`.tts_request(endpoint, method, body, expect_binary)`

**Arguments**

- `endpoint`            Character. The API endpoint (e.g., `"v1/audio/speech"`).
- `method`             Character. HTTP method (`"GET"` or `"POST"`).
- `body`                List or NULL. Request body for POST requests.
- `expect_binary`      Logical. If TRUE, expect binary response (audio data).

**Value**

Response content (raw bytes if binary, parsed JSON otherwise).

---

|                                   |                                                 |
|-----------------------------------|-------------------------------------------------|
| <code>chatterbox_available</code> | <i>Check if Chatterbox Service is Available</i> |
|-----------------------------------|-------------------------------------------------|

---

**Description**

Quick check if the local Chatterbox TTS API is reachable. Returns TRUE/FALSE without throwing errors.

**Usage**

`chatterbox_available(port, timeout)`

**Arguments**

- `port`                Port to check (default from `CHATTERBOX_PORT` env var or 4123)
- `timeout`            Timeout in seconds (default 2)

**Value**

TRUE if service is available, FALSE otherwise

## Examples

```
## Not run:
  if (chatterbox_available()) {
    speech("Hello", voice = "default", backend = "chatterbox")
  }

## End(Not run)
```

---

chatterbox\_voice\_upload

*Upload Voice to Chatterbox*

---

## Description

Uploads a voice sample directly to a local Chatterbox instance. Does not require setting the API base URL first.

## Usage

```
chatterbox_voice_upload(voice_file, voice_name, port, timeout)
```

## Arguments

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| voice_file | Character. Path to the voice sample file (mp3, wav).                   |
| voice_name | Character. Name to save the voice as.                                  |
| port       | Port for Chatterbox API (default from CHATTERBOX_PORT env var or 4123) |
| timeout    | Timeout in seconds (default 30)                                        |

## Value

TRUE if upload succeeded, FALSE otherwise. Does not throw errors.

## Examples

```
## Not run:
  # Upload a voice sample
  success <- chatterbox_voice_upload("my_voice.wav", "my-voice")

  if (success) {
    speech("Hello!", voice = "my-voice", backend = "chatterbox")
  }

## End(Not run)
```

---

|                   |                               |
|-------------------|-------------------------------|
| elevenlabs_voices | <i>List ElevenLabs Voices</i> |
|-------------------|-------------------------------|

---

**Description**

Get all voices available in your ElevenLabs account.

**Usage**

```
elevenlabs_voices()
```

**Value**

Data frame with voice\_id, name, and category columns.

**Examples**

```
## Not run:
voices <- elevenlabs_voices()
print(voices)

## End(Not run)
```

---

|                         |                                |
|-------------------------|--------------------------------|
| elevenlabs_voice_delete | <i>Delete ElevenLabs Voice</i> |
|-------------------------|--------------------------------|

---

**Description**

Delete a voice from your ElevenLabs account.

**Usage**

```
elevenlabs_voice_delete(voice_id)
```

**Arguments**

voice\_id            Character. The voice ID to delete.

**Value**

Invisible TRUE on success.

**Examples**

```
## Not run:
elevenlabs_voice_delete("abc123")

## End(Not run)
```



---

`elevenlabs_voice_upload`*Upload Voice to ElevenLabs (Instant Voice Clone)*

---

**Description**

Create an instant voice clone on ElevenLabs from audio samples.

**Usage**

```
elevenlabs_voice_upload(files, name, description, remove_background_noise, labels)
```

**Arguments**

|                                      |                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------|
| <code>files</code>                   | Character vector. Paths to audio files (1-25 files, each up to 10MB).               |
| <code>name</code>                    | Character. Name for the cloned voice.                                               |
| <code>description</code>             | Character or NULL. Optional description.                                            |
| <code>remove_background_noise</code> | Logical. Remove background noise from samples. Default FALSE.                       |
| <code>labels</code>                  | Named list or NULL. Optional labels (e.g., <code>list(accent = "british")</code> ). |

**Value**

List with `voice_id` and `name` on success.

**Examples**

```
## Not run:
# Clone from a single file
voice <- elevenlabs_voice_upload(
  files = "my_voice.mp3",
  name = "My Clone"
)

# Use the cloned voice
speech("Hello!", voice = voice$voice_id, backend = "elevenlabs")

# Clone from multiple samples
voice <- elevenlabs_voice_upload(
  files = c("sample1.mp3", "sample2.mp3"),
  name = "Better Clone",
  remove_background_noise = TRUE
)

## End(Not run)
```

---

|           |                                 |
|-----------|---------------------------------|
| languages | <i>List Supported Languages</i> |
|-----------|---------------------------------|

---

**Description**

Returns a list of languages supported by the TTS server.

**Usage**

```
languages()
```

**Value**

A character vector or list of supported language codes.

**Examples**

```
## Not run:  
set_tts_base("http://localhost:4123")  
langs <- languages()  
print(langs)  
  
## End(Not run)
```

---

|                    |                               |
|--------------------|-------------------------------|
| set_elevenlabs_key | <i>Set ElevenLabs API Key</i> |
|--------------------|-------------------------------|

---

**Description**

Configure the API key for ElevenLabs TTS backend.

**Usage**

```
set_elevenlabs_key(api_key)
```

**Arguments**

|         |                                     |
|---------|-------------------------------------|
| api_key | Character. Your ElevenLabs API key. |
|---------|-------------------------------------|

**Examples**

```
## Not run:  
set_elevenlabs_key("your-api-key-here")  
# Or use environment variable ELEVENLABS_API_KEY  
  
## End(Not run)
```

---

|              |                                 |
|--------------|---------------------------------|
| set_tts_base | <i>Set the TTS API Base URL</i> |
|--------------|---------------------------------|

---

**Description**

Sets the base URL for the TTS API. This should be the root URL without trailing slash (e.g., "http://localhost:4123" or "https://api.openai.com").

**Usage**

```
set_tts_base(url)
```

**Arguments**

|     |                                             |
|-----|---------------------------------------------|
| url | Character string. The base URL for the API. |
|-----|---------------------------------------------|

**Value**

Invisibly returns the previous value.

**Examples**

```
## Not run:  
# For local Chatterbox server  
set_tts_base("http://localhost:4123")  
  
# For OpenAI  
set_tts_base("https://api.openai.com")  
  
## End(Not run)
```

---

|             |                            |
|-------------|----------------------------|
| set_tts_key | <i>Set the TTS API Key</i> |
|-------------|----------------------------|

---

**Description**

Sets the API key for authentication. Required for cloud services like OpenAI, typically ignored by local servers.

**Usage**

```
set_tts_key(key)
```

**Arguments**

|     |                                |
|-----|--------------------------------|
| key | Character string. The API key. |
|-----|--------------------------------|

**Value**

Invisibly returns the previous value.

**Examples**

```
## Not run:
set_tts_key(Sys.getenv("OPENAI_API_KEY"))

## End(Not run)
```

---

|        |                                  |
|--------|----------------------------------|
| speech | <i>Generate Speech from Text</i> |
|--------|----------------------------------|

---

**Description**

Generate audio from text using various TTS backends: local Chatterbox server, OpenAI TTS API, or ElevenLabs API.

**Usage**

```
speech(input, voice, file = NULL, backend = c("auto", "chatterbox",
  "openai", "elevenlabs"), model = NULL, temperature = NULL,
  speed = NULL, exaggeration = NULL, cfg_weight = NULL,
  stability = NULL, similarity_boost = NULL, seed = NULL,
  response_format = NULL, instructions = NULL)
```

**Arguments**

|              |                                                                                                                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input        | Character. The text to convert to speech.                                                                                                                                                                                            |
| voice        | Character. The voice to use for synthesis. For OpenAI: "alloy", "echo", "fable", "onyx", "nova", "shimmer". For ElevenLabs: voice ID (e.g., "XpDLYThV0yUAFjVTok7m"). For Chatterbox: custom voice names uploaded via voice_upload(). |
| file         | Character or NULL. Output file path. If NULL, returns raw bytes.                                                                                                                                                                     |
| backend      | Character. Backend to use: "chatterbox" for local server, "openai" for OpenAI TTS API, "elevenlabs" for ElevenLabs API, or "auto" to use configured API base.                                                                        |
| model        | Character or NULL. The model to use. For OpenAI: "tts-1" or "tts-1-hd". For ElevenLabs: "eleven_multilingual_v2" (default), "eleven_turbo_v2_5", etc. For Chatterbox: optional, often ignored.                                       |
| temperature  | Numeric or NULL. Sampling temperature for generation.                                                                                                                                                                                |
| speed        | Numeric or NULL. Speed multiplier for the audio.                                                                                                                                                                                     |
| exaggeration | Numeric or NULL. Exaggeration parameter (Chatterbox-specific).                                                                                                                                                                       |
| cfg_weight   | Numeric or NULL. CFG weight parameter (Chatterbox-specific).                                                                                                                                                                         |
| stability    | Numeric or NULL. Voice stability 0-1 (ElevenLabs-specific). Default 0.5.                                                                                                                                                             |

|                  |                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| similarity_boost | Numeric or NULL. Similarity boost 0-1 (ElevenLabs-specific). Default 0.75.                                                        |
| seed             | Integer or NULL. Random seed for reproducible output.                                                                             |
| response_format  | Character or NULL. Audio format (e.g., "wav", "mp3"). If NULL and file is provided, inferred from file extension.                 |
| instructions     | Character or NULL. Instructions for how the voice should speak (OpenAI-specific, e.g., "Speak in a cheerful and positive tone."). |

**Value**

If `{file}` is provided, invisibly returns the file path. If `{file}` is NULL, returns raw audio bytes.

**Examples**

```
## Not run:
# Using local Chatterbox server
set_tts_base("http://localhost:4123")
speech("Hello, world!", voice = "FatherChristmas", file = "hello.wav")

# Using OpenAI TTS
speech("Hello, world!", voice = "nova", file = "hello.mp3", backend = "openai")

# Using ElevenLabs
speech("Hello, world!", voice = "XpDLYThV0yUAFjVTok7m",
      file = "hello.mp3", backend = "elevenlabs")

## End(Not run)
```

speech\_clone

*Generate Speech with Voice Cloning***Description**

Uploads a voice sample and generates speech in that voice using the `/v1/audio/speech/upload` endpoint.

**Usage**

```
speech_clone(input, voice_file, file, exaggeration, temperature, cfg_weight, speed, seed)
```

**Arguments**

|            |                                                                  |
|------------|------------------------------------------------------------------|
| input      | Character. The text to convert to speech.                        |
| voice_file | Character. Path to the voice sample file (mp3, wav, etc.).       |
| file       | Character or NULL. Output file path. If NULL, returns raw bytes. |

|              |                                                                |
|--------------|----------------------------------------------------------------|
| exaggeration | Numeric or NULL. Exaggeration parameter (Chatterbox-specific). |
| temperature  | Numeric or NULL. Sampling temperature for generation.          |
| cfg_weight   | Numeric or NULL. CFG weight parameter (Chatterbox-specific).   |
| speed        | Numeric or NULL. Speed multiplier for the audio.               |
| seed         | Integer or NULL. Random seed for reproducible output.          |

**Value**

If `{file}` is provided, invisibly returns the file path. If `{file}` is NULL, returns raw audio bytes.

**Examples**

```
## Not run:
set_tts_base("http://localhost:4123")

# Clone voice and generate speech
speech_clone(
  input = "Hello with my custom voice!",
  voice_file = "my_voice.mp3",
  file = "output.wav",
  exaggeration = 0.8
)

## End(Not run)
```

---

|            |                             |
|------------|-----------------------------|
| tts_health | <i>Check TTS API Health</i> |
|------------|-----------------------------|

---

**Description**

Checks whether the TTS backend is reachable by trying common health endpoints.

**Usage**

```
tts_health()
```

**Value**

A list with components: `\describe{ \item{ok}{Logical. TRUE if the server is reachable.} \item{status}{Character. A human-readable status message.} \item{raw}{The raw response from the server (if any).} }`

**Examples**

```
## Not run:
set_tts_base("http://localhost:4123")
h <- tts_health()
if (h$ok) {
  message("Server is ready!")
}

## End(Not run)
```

---

|               |                                    |
|---------------|------------------------------------|
| tts_providers | <i>TTS Provider Configurations</i> |
|---------------|------------------------------------|

---

**Description**

A list of known TTS providers with their default voices and configuration.

**Details**

A named list where each element contains: \describe{ \item{ voices } { Character vector of default/known voice names, or NULL if dynamic} \item{ env\_var } { Name of environment variable for API key, or NULL if not needed} \item{ base\_url } { Default API base URL, or NULL } }

**Examples**

```
names(tts_providers)
tts_providers[["OpenAI"]]$voices
```

---

|            |                                      |
|------------|--------------------------------------|
| tts_voices | <i>Get Voices for a TTS Provider</i> |
|------------|--------------------------------------|

---

**Description**

Returns available voices for a given provider. Attempts to fetch dynamically from the API first, falls back to static list if unavailable.

**Usage**

```
tts_voices(provider, base_url, timeout)
```

**Arguments**

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| provider | Character string naming the provider (e.g., "OpenAI", "Chatterbox (Local)") |
| base_url | Optional base URL override for the provider's API                           |
| timeout  | Timeout in seconds for API requests (default 2)                             |

**Value**

Character vector of voice names

**Examples**

```
## Not run:
tts_voices("OpenAI")
tts_voices("Chatterbox (Local)")

## End(Not run)
```

---

voices

*List Available Voices*

---

**Description**

Returns a list of available voices from the TTS server. Tries the /v1/audio/voices endpoint first, then falls back to /voices.

**Usage**

```
voices()
```

**Value**

A character vector of voice names, or a list/data.frame depending on the server's response format.

**Examples**

```
## Not run:
set_tts_base("http://localhost:4123")
v <- voices()
print(v)

## End(Not run)
```

---

voice\_upload

*Upload Voice to Library*

---

**Description**

Uploads a voice sample to the server's voice library for reuse. Once uploaded, the voice can be used by name in speech().

**Usage**

```
voice_upload(voice_file, voice_name, language)
```



**Arguments**

|            |                                                            |
|------------|------------------------------------------------------------|
| voice_file | Character. Path to the voice sample file (mp3, wav, etc.). |
| voice_name | Character. Name to save the voice as.                      |
| language   | Character or NULL. Language code (e.g., "en", "fr", "es"). |

**Value**

Invisibly returns the response from the server.

**Examples**

```
## Not run:
set_tts_base("http://localhost:4123")

# Upload a voice
voice_upload(
  voice_file = "my_voice.wav",
  voice_name = "my-custom-voice"
)

# Upload with language
voice_upload(
  voice_file = "french_voice.wav",
  voice_name = "french-speaker",
  language = "fr"
)

# Then use it by name
speech(
  input = "Hello with my custom voice!",
  voice = "my-custom-voice",
  file = "output.wav"
)

## End(Not run)
```

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